# Computational Complexity

Reading guide and Homework for **Week 03**

## 1 Reading Guide

In the fourth class:

- We proved the Cook-Levin Theorem, that 3SAT is NP-complete. Read §2.3.
- We showed the NP-hardness of CLIQUE/INDSET, directed hamiltonian path, integer linear programming, and quadratic programming. Read §2.4.
- We defined NP using non-deterministic Turing machines. Read §2.1.2.

There were several details that we covered very quickly, and which **must** be looked at carefully, in order to properly understand what is going on. This kind of familiarity will be necessary to understand what's being said in class.

## 2 Homework Exercises

Exercises are organized by the skill that is being trained. The test and exam questions will always be about testing one of the skills appearing in the homework.

**Skill:** *show that a given problem is* NP*-complete. This is one of the most fundamental skills taught in the course. Become a master in it.*

1. Look at figure 2.4 in the book (Section 2.4, page 51). It shows a graph of reductions between NP-complete problems. Solve every exercise in this figure (exs 2.15 to 2.22)

**Skill:** *Learn new concepts by yourself. (The test will have questions about these, possibly asking you to prove something that appears in these sections, or reason in a way that requires that you understood them.).*

2. Decision vs. Search. We talk about NP as being a class of *decision problems*. But we could also consider the equivalent *search problems*. So we may wish to *output* a satisfying assignment to a CNF formula, instead of just deciding whether such an assigment exists. It turns out, for NP-complete problems, these two tasks have the same difficulty. Read and understand §2.5.
3. We have defined NP using non-deterministic turing machines. They also obey a time-hierarchy theorem. Read and understand §3.2.